Unnamed Local Roads on Tribal Reservations: A Coverage Gap the Bias Scorecard Cannot See

Craig Carlos Ouma*[1]

[1]Independent Researcher

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Abstract

The challenge's coverage measures and every Bias Scorecard metric count road length and features, so a road drawn without a name counts as fully covered. Using the challenge data (Overture Maps release 2026-08-19.0 and the strata tables), this entry measures the share of drivable road length with no name in each of 9,794 scored tracts and compares tribal with non-tribal tracts within the same region and rurality class. On 80 federally recognised reservation tracts, local streets are unnamed 9.5 percentage points more often than on matched non-tribal tracts (95 % county bootstrap interval 3.5 to 14.7), while highways on the same tracts show no difference (−0.4). The Osage Reservation reaches +27.6 points, with 57 % of its drivable length unnamed. In 19 reservation tracts, home to 55,092 people, at least 30 % of local street length has no name, against 13 % of comparable non-tribal tracts. Several of these tracts combine unnamed streets with recent wildfire and high social vulnerability. Emergency dispatch, evacuation orders and relief operations locate people by street name, so these residents are the hardest to reach in exactly the events the challenge regions face.

1 The pattern and why the scorecard cannot see it

Every coverage measure in this challenge, and every metric on the Bias Scorecard, counts length or features. A road segment that Overture holds without a name adds as much length as a named one, so a tract whose roads are all drawn but half unnamed scores as fully covered. The name is what emergency services use. A caller gives an address, computer-aided dispatch validates it against road centrelines that carry street names and address ranges [3], and routing tools give instructions by street name. This entry shows that on federally recognised reservations a much larger share of the local street network has no name in Overture than on comparable non-tribal tracts, while the highways through the same reservations are named as often as anywhere else.

2 Measure and comparison

For each scored tract the measure is the share of drivable road length (Overture [2] `roads` layer, classes motorway to unclassified, clipped to the tract) whose `names.primary` is empty. It is computed separately for the local classes (tertiary, residential, unclassified) and the named classes

*Correspondence: `craigcarlos95@gmail.com`. Zindi: `craigouma` (solo). Leaderboard submissions tied to the methodology: S25 (`2ncgdcQQ`) and S24 (`EC3xBxjr`). Entry for Best Bias Discovery, Bias Bounty Mapping Equity Challenge. The full methodology is submitted separately for Best Documentation.

(motorway, trunk, primary, secondary). Tracts are tribal when the challenge's tribal overlap table links them to an American Indian, Alaska Native or Native Hawaiian area, split into legal areas (reservations and off-reservation trust land) and statistical areas (Oklahoma Tribal Statistical Areas, OTSAs).

Rurality changes naming a great deal: in South-Central Texas the unnamed share is 8 % in metropolitan tracts and 29 % in rural ones. Tribal tracts are therefore compared with non-tribal tracts in the same region and RUCA class [5] (metropolitan, micropolitan, small town, rural), and the cell differences are averaged with weights equal to the number of tribal tracts in each cell. Intervals come from 1,000 bootstrap resamples of whole counties, which keeps neighbouring tracts that share mapping history together. All numbers come from stage 04 run `20261030-1019-04_analysis-8a838bb` on the challenge data only; county names come from the Census 2020 county code list [6].

Reproducing the finding. Road length comes from `<region>-overture-roads.parquet` (classes motorway, trunk, primary, secondary, tertiary, residential, unclassified), as geodesic metres of each segment clipped to the tract polygons in `<region>-census-tracts.parquet`; a segment is unnamed when `names.primary` is null. The tribal kind comes from `<region>-strata-tract-table.parquet`: a tract is a reservation tract when `tribal_legal` is true and a statistical-area tract when `tribal_any` is true and `tribal_legal` is false; area names come from `aiannh_name` in `<region>-tribal-tract-table.parquet`. The rurality class maps `ruca_primary` 1 to 3 to metropolitan, 4 to 6 to micropolitan, 7 to 9 to small town and 10 to rural. Tracts with no drivable road are left out, and tracts with no local road are left out of the local comparison. In each region and class cell holding both groups, the mean tribal share minus the mean non-tribal share is taken, and the cells are averaged with weights equal to their tribal tract counts. The interval resamples counties (the first five GEOID digits) with replacement 1,000 times and reports the 2.5th and 97.5th percentiles, dropping resamples in which no cell holds both groups.

3 Evidence

Figure 2 maps the measure. Table 1 and Figure 1 give the matched differences. On the 80 reservation tracts, local streets are unnamed 9.5 percentage points more often than on matched non-tribal tracts (95 % interval 3.5 to 14.7). The highways on the same tracts show no difference (−0.4, interval −7.3 to 5.6), so the gap sits on the streets where people live and the measure is not picking up a regional habit of leaving highways unnamed. The difference holds in each region with enough reservation tracts to test: Maricopa (37 tracts, +7.3), Eastern Washington (16, +4.5) and Eastern Oklahoma, where the Osage Reservation reaches +27.6 (9.8 to 29.6). Northern California's 13 reservation tracts point the same way with an interval that includes zero.

The OTSAs hold the largest unnamed networks in absolute terms: 13,066 km of unnamed drivable road in the Cheyenne and Arapaho OTSA (47 % of its network) and 8,831 km in the Choctaw OTSA (34 %). After matching they do not separate from non-tribal tracts, because 792 of Eastern Oklahoma's 1,192 scored tracts lie in an OTSA and only 43 non-tribal tracts outside its metropolitan areas remain to compare with. The OTSA result is therefore unresolved by this design, and the claim made here is limited to reservations.

The 80 reservation tracts hold 294,612 residents and 8,266 km of unnamed drivable road. In 19 of them, home to 55,092 people, at least 30 % of local street length has no name. That is 24 % of reservation tracts, against 13 % of non-tribal tracts once those are weighted to the same mix of rurality.

Table 1: Tribal minus non-tribal unnamed share of drivable length, matched within region and RUCA class, in percentage points, with 95 % county bootstrap intervals.

Tribal tracts	Roads	Tracts	Difference	Interval
Reservations, all regions	local	80	+9.5	3.5 to 14.7
Reservations, all regions	all drivable	80	+8.3	2.9 to 13.5
Reservations, all regions	named classes	79	−0.4	−7.3 to 5.6
Osage Reservation, Eastern Oklahoma	local	13	+27.6	9.8 to 29.6
Reservations, Maricopa	local	37	+7.3	3.0 to 10.9
Reservations, Eastern Washington	local	16	+4.5	0.0 to 14.8
Reservations, Northern California	local	13	+3.8	−2.9 to 10.1
OTSAs, Eastern Oklahoma	local	792	+3.0	−6.0 to 7.5

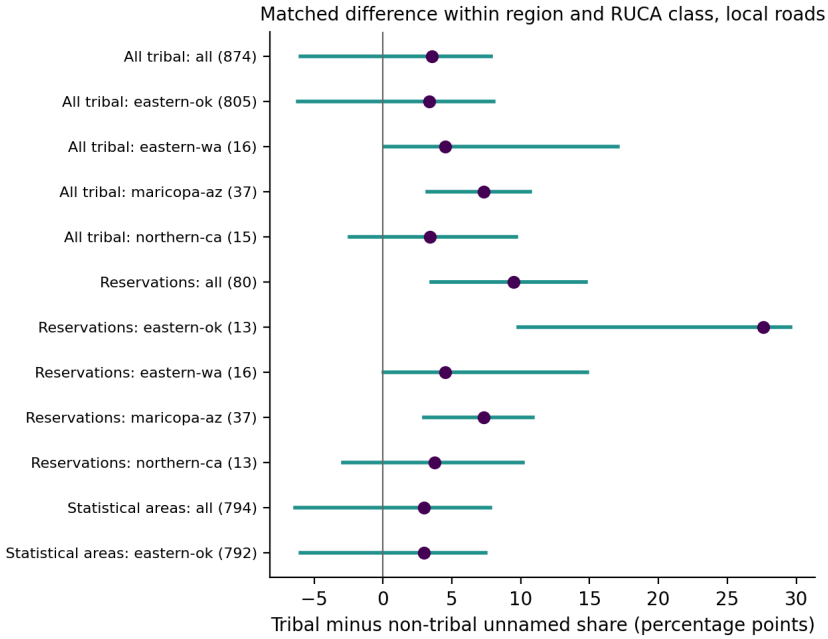


Figure 1: Matched difference in the unnamed share of local street length, tribal minus non-tribal tracts, with 95 % county bootstrap intervals. Numbers in brackets are tribal tracts.

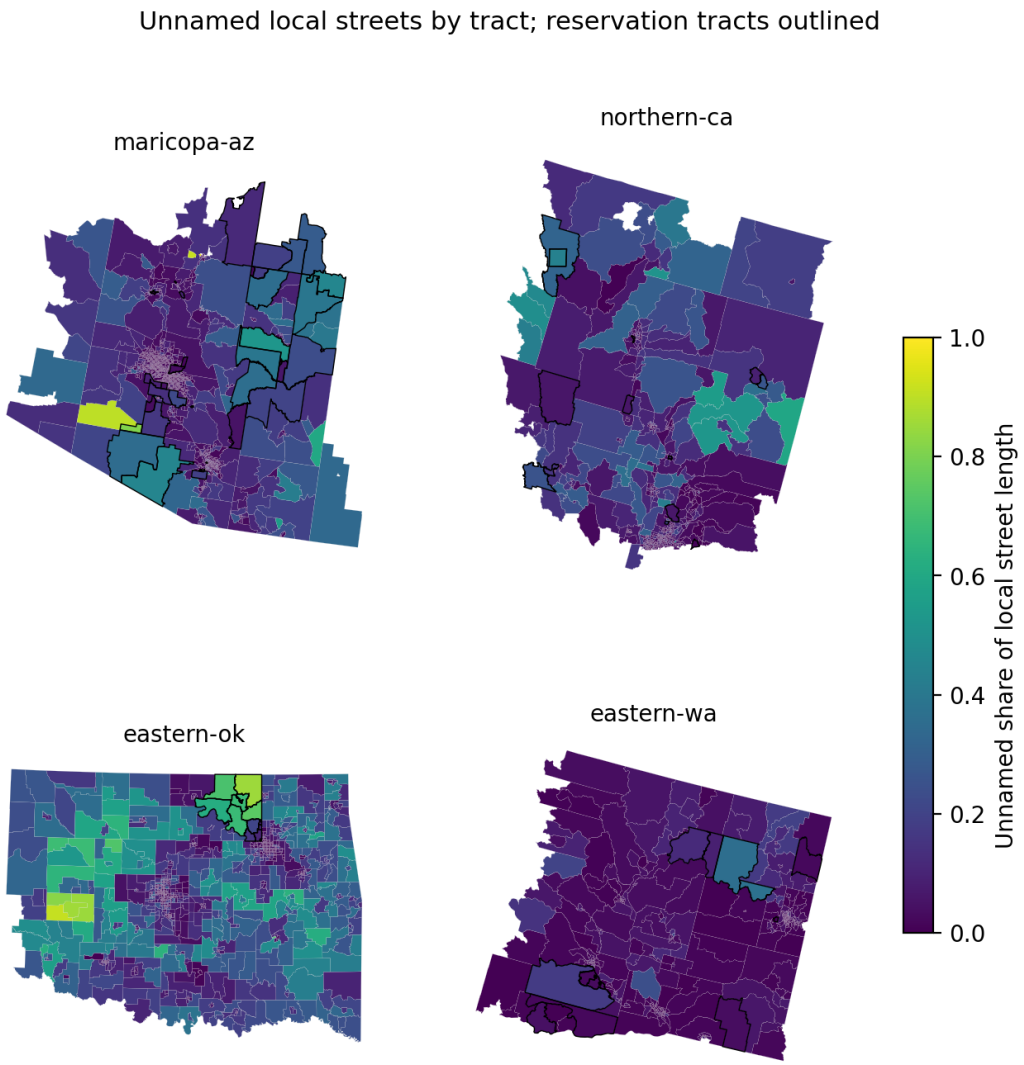


Figure 2: Unnamed share of local street length by tract (Overture 2026-08-19.0). Reservation tracts are outlined in black; the Osage Reservation is the outlined block in north-central Oklahoma. Oklahoma Tribal Statistical Areas are not outlined. Grey tracts have no local road.

Table 2 names the reservation tracts with the most unnamed road, with their wildfire history and social vulnerability from the challenge's strata tables. Several of them combine a largely unnamed street network with recent fire and high vulnerability. Tract 9400.04 in Osage County, Oklahoma has 78 % of its drivable length unnamed and 29 % of its land burned between 2015 and 2023. Tract 9400 in Ferry County, Washington, on the Colville Reservation, has 498 km of unnamed road, 27 % of its land burned since 2015 and an SVI percentile of 0.84. Tracts 9408 and 9411 in Pima County, Arizona, on the Tohono O'odham Nation Reservation in the Sonoran Desert, have SVI percentiles of 0.96 and a third or more of local street length unnamed.

Table 2: Reservation tracts with the most unnamed drivable road. Local: unnamed share of local street length. Burned: share of land burned 2015 to 2023 (MTBS). SVI: CDC SVI 2022 overall percentile [4].

GEOID	Tract	County	Reservation	Unnamed km	All	Local	Burned	SVI
40113940004	9400.04	Osage, OK	Osage	606	0.78	0.86	0.29	0.48
40113940003	9400.03	Osage, OK	Osage	524	0.62	0.62	0.12	0.74
53019940000	9400	Ferry, WA	Colville	498	0.34	0.36	0.27	0.84
04001970202	9702.02	Apache, AZ	Zuni	462	0.35	0.39	0.00	0.54
40113940011	9400.11	Osage, OK	Osage	454	0.74	0.71	0.12	0.58
06023010102	101.02	Humboldt, CA	Yurok	387	0.32	0.32	0.20	0.61
04001970502	9705.02	Apache, AZ	Fort Apache	385	0.23	0.23	0.02	0.66
04019940800	9408	Pima, AZ	Tohono O'odham	384	0.40	0.45	0.03	0.96
40113940001	9400.01	Osage, OK	Osage	327	0.61	0.68	0.37	0.83
04017940100	9401	Navajo, AZ	Fort Apache	324	0.47	0.52	0.07	0.87

4 Consequences

Emergency dispatch. An address on a road with no name in the map cannot be matched to a centreline, so a 911 call from it geocodes to a nearby named road, a ZIP centroid or nothing. On a reservation, response is often shared between tribal, county, state and federal agencies, and units arriving from outside lack the local knowledge that substitutes for a street name. On the Osage Reservation, where 57 % of the drivable length has no name and single tracts reach 78 %, a crew from a neighbouring county routed by a tool built on open map data receives instructions in which most turns are onto an unnamed road.

Evacuation routing. Wildfire evacuation orders name the roads to leave by and the zones they bound. Where a third of the local network is unnamed, as on the Colville and Yurok tracts with a fifth to a quarter of their land burned since 2015, zone maps and routing products drawn from open data cannot describe the egress routes that residents of those roads would need.

Disaster relief. Damage assessment, supply distribution and welfare checks during extreme heat, which the Tohono O'odham tracts face each summer, are organised by address. Households on unnamed roads drop out of address-based lists and have to be found by coordinates.

Why the scorecard misses it. The Road Network Equity Ratio and the other scorecard metrics compare lengths and counts, and the transport component itself compares only motorway to secondary length. A reservation whose streets are drawn but unnamed has full road coverage by

every one of these measures. The disparity becomes visible only through an attribute, the name, that none of them reads.

5 Reproducibility

Every number comes from stage 04 run `20261030-1019-04_analysis-8a838bb` of the challenge pipeline (seed 20261025, 1,000 county bootstrap resamples per interval), which read only the challenge bucket [1] and the Census 2020 county code list [6], retrieved 29 October 2026, for county names. Runs `20261029-2348-04_analysis-503f5c5` and `20261030-1019-04_analysis-8a838bb` of the same analysis code produced identical numbers; the maps come from the second. The tract-level tables (`unnamed_worst_reservation_tracts.csv`, `unnamed_by_tribal_area.csv`) are produced by the same run. The measure reads Overture names as published. A road may carry a name in a local addressing system that Overture does not hold; in that case the open map lacks the name, which is the gap being measured.

References

[1] Humane Intelligence (2026). Bias Bounty Mapping Equity Challenge data repository and README. Source Cooperative. `https://source.coop/humane-intelligence/bias-bounty-mapping-equity-challenge`

[2] Overture Maps Foundation (2026). Overture Maps release 2026-08-19.0 and schema documentation. `https://docs.overturemaps.org/`

[3] National Emergency Number Association (2022). NENA Standard for NG9-1-1 GIS Data Model, NENA-STA-006.2-2022.

[4] Centers for Disease Control and Prevention and ATSDR (2024). CDC/ATSDR Social Vulnerability Index 2022.

[5] USDA Economic Research Service (2025). Rural-Urban Commuting Area codes, 2020.

[6] US Census Bureau. 2020 national county code list (`national_county2020.txt`). Retrieved 29 October 2026. `https://www2.census.gov/geo/docs/reference/codes2020/national_county2020.txt`